

Schemas in Weights, Facts in the Organizer: A Controlled Test of Fact Offloading in Small Language Model Pretraining

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Abstract

Small language models are widely believed to waste parametric capacity on factual memorization. The phi-3 team filters facts out of training data explicitly to preserve capacity for reasoning, and recent memory-augmented architectures are motivated by the same intuition. Yet no controlled experiment has measured whether offloading facts during pretraining actually changes what the remaining capacity learns. We present the design and pilot results of the first such experiment. Twin decoder models train on byte-matched corpora under identical schedules; the only difference is that the split arm sees every fact value wrapped in database lookup calls whose value tokens are masked from the loss, so it is never rewarded for memorizing a fact, only for asking a minimal external key-value store we call the organizer. Because we generate the facts ourselves, the fact load is a controlled, swept variable with known information content. Pilot runs at 160M parameters establish the mechanism decisively and repeatedly: the split model stores zero measured fact bits in its weights, answers 99.8 percent of fact probes with the organizer attached and none without it, composes correct lookups for entities it has never seen in any training document (at or near 100 percent across rounds), and roughly doubles its dense twin on held-out fact-use question answering (56.9 to 70.6 percent versus 27.1 to 32.9 percent) at identical parameter count, data content, and data order. The pilots also produced a useful negative finding: knowledge-free symbolic reasoning tasks did not rise above chance from in-mixture exposure at pilot budgets, across three rounds of mixture and difficulty remediation, which motivated a preregistered re-anchoring of the primary endpoint to fact-use reasoning before any confirmatory run began. The full preregistered battery, a dose-response sweep at 160M and a billion-parameter paired confirmation, is running now on frozen margins and decision rules. We argue the design closes the gap between a widely deployed intuition and its first measurement, and that the pilot evidence already justifies scaling the paradigm.

1 Introduction

A small language model has a fixed budget of parameters, and everything it learns must share that budget. Facts about the world, which are numerous, arbitrary, and individually low-value, compete for storage with the reusable procedures that make a model useful: parsing, composing, comparing, deducing. This tension is not hypothetical. The phi-3 technical report describes deleting fact-heavy web pages from training data precisely to “leave more model capacity for reasoning for the mini size models” [1]. A production model family was designed around the belief that facts crowd out reasoning in small models.

That belief has never been tested directly. Data filtering, as in phi-3, confounds fact removal with every other property of the removed documents. Retrieval augmentation at inference time, the standard industrial remedy, leaves the memorization pressure of pretraining fully intact, and a recent systematic study finds that it leaves reasoning performance essentially unchanged across model scales [3]. The Limited Memory Language Model (LMLM) line of work [2] built the missing training-time mechanism, pretraining models that emit database lookup calls with the retrieved values masked from the loss, and showed striking gains in factual precision per parameter. On the question of whether the freed capacity buys anything, however, the LMLM paper reports only parity on five natural language understanding benchmarks and states in its limitations that benchmarks disentangling knowledge from reasoning “remain underexplored.” The conjecture at the center of the small-model design space is, in short, open.

This paper describes an experiment built to close it, and reports the evidence from its pilot phase. The design principle throughout is that the corpus itself is the measuring instrument. We generate the facts, so we know exactly how many bits of factual content the dense arm is being asked to store, we can sweep that load as a controlled dose, and we can annotate the split arm’s lookup calls with perfect coverage rather than with a learned annotator. We generate the reasoning tasks, so held-out accuracy is scored by construction-time oracles rather than heuristics, and contamination is excluded by structure hashing rather than by n-gram guesswork. Both arms of every pair see byte-identical natural text, identical synthetic content, identical document order, and matched per-component token budgets, so the contrast isolates a single bit of information: whether the loss function rewards memorizing fact values.

Our contributions are threefold. First, a controlled experimental design, to our knowledge the first, that combines training-time fact offloading, a matched-pair baseline, a swept fact dose with known bit content, and reasoning as a preregistered primary endpoint. Second, pilot-scale evidence that the mechanism does exactly what the hypothesis requires, replicated across three independent training rounds. Third, a preregistered confirmatory battery, frozen before any confirmatory run started, currently executing at 160M and one billion parameters.

2 Related work

Training with external memory. LMLM [2] pretrains GPT-2 and Llama style models from scratch with entity facts wrapped in lookup calls and masked from the loss, using a fine-tuned annotator model to insert calls over Wikipedia text and a fuzzy retriever over 54.6M mined triplets at inference. A 382M LMLM matches the factual precision of a 7B baseline, and unplugging the database collapses it, establishing that facts genuinely live outside the weights. We adopt this training mechanism directly and gratefully. What LMLM does not do is measure a reasoning consequence; its general-capability evaluation is five benchmarks near their random floors at that scale, reported as parity. RETRO [5] trains with chunked cross-attention into a trillion-token store and gains mainly on copying-adjacent perplexity and factual QA; its own analysis attributes much of the gain to overlap exploitation. kNN-LM style post-hoc memory actively degrades reasoning even with oracle retrieval [6]. Memory layers [7] and related sparse in-weight stores relocate memorization inside the gradient loop; they test architectural separation, not external offloading, and their stores cannot be edited or unplugged.

Measurements adjacent to the hypothesis. A 2026 scaling study of retrieval-considerate pretraining [3] trades corpus size against datastore size and finds retrieval substitutes for parameters on knowledge tasks while reasoning tasks show minimal change; retrieval there is inference-time, so weights still pay the memorization cost during training. Charpentier et al. [4] train small masked

LMs with idealized retrieval and observe reallocation toward syntax with a regression in broader understanding, a caution that freed capacity does not automatically become the capability one wants. Allen-Zhu and Li quantify parametric fact storage at roughly two bits per parameter under sufficient exposure [8] and supply the accounting method our mediation analysis adapts, as well as the synthetic-task methodology our reasoning corpora follow [9].

Theory. Complementary learning systems theory [10, 11] holds that intelligent agents need a fast, item-specific store alongside a slow structure learner, because arbitrary item memory interferes with the extraction of generalizable structure. Our experiment operationalizes the division of labor directly: the organizer plays the fast store, the network is freed to be the slow learner, and the theory’s signature predictions (division of labor, reduced interference, faster schema acquisition) map onto our H1 through H4.

3 Method

3.1 Two arms, one difference

Every condition trains a pair of decoder-only transformers (RMSNorm, rotary positions, SwiGLU, untied embeddings; 12 layers by 768 for the 160M class, 22 by 1792 for the 1B class) with the GPT-2 byte-pair tokenizer extended by four special tokens. Members of a pair share everything: initialization seed, data-order seed, batch schedule, learning rate schedule, corpus content, document order, and per-component token budgets matched within one percent. They differ in exactly one respect. In the dense arm a biography sentence reads

Kai Nakamura majored in Communications at ...

and every token carries loss. In the split arm the same sentence reads

Kai Nakamura majored in <|db_start|>Kai Nakamura, major<|db_retrieve|> Communications<|db_end|>
at ...

and the value tokens between the retrieve and end markers are excluded from the loss. The split model conditions on values in context but receives no gradient reward for predicting them. It is graded on knowing when to open a call and how to compose the query. Nothing is retrieved during training; the values sit inline, masked, so the training loop is standard next-token prediction plus a label mask, with zero added FLOPs.

At inference the organizer comes alive. It is deliberately minimal: an exact-match table from normalized (**entity**, **relation**) keys to value strings, built by the corpus generator. A three-state interception loop in the decoder watches for <|db_start|>, records the query the model composes, executes the lookup at <|db_retrieve|>, and force-decodes the returned value and closing marker through the model’s cache before generation resumes. Hits, misses, and malformed calls are all counted. Using exact match over canonical synthetic keys removes retriever quality as a confound entirely: if the split arm wins, no reviewer can attribute the win to a clever retriever, because the store is trivially perfect and its behavior fully audited.

3.2 The corpus as instrument

Both arms train on a five-component mixture (final frozen recipe: 54 percent natural web text from FineWeb-Edu, 23 percent synthetic biographies, 12 percent synthetic mathematics, 8 percent synthetic deduction, 3 percent fact-use question answering). The biographies are the dose. Each

of N entities carries six attributes drawn from finite pools with a combined information content of about 53 bits per entity, rendered through more than twenty paraphrase templates per attribute, eight sentence orderings, and three name forms. Sweeping N (50k, 200k, 800k at 160M scale; 800k and 4M at 1B scale) sweeps the number of facts the dense arm must store while the split arm may ignore. Because the generator knows every fact, annotation coverage in the split arm is exact by construction, unlike learned annotation pipelines which leak unwrapped facts into ordinary text.

The reasoning components are knowledge-free by design. The mathematics family follows the iGSM methodology [9]: random dependency graphs over nonsense quantities (“crimson satchels in the Annex”), arithmetic modulo 23 so no memorized number facts help, chain-of-thought solutions, difficulty parameterized by operation count, and an independent oracle that re-parses each prompt and recomputes the answer. The deduction family poses Horn-clause chains over nonsense predicates with balanced yes and no labels and a forward-chaining oracle. Fact-use QA asks questions whose chain of thought must cite stored facts and then reason over them: date comparisons, city equality checks, and extractions. Held-out sets are disjoint by structure hash, and a separate probe set contains fresh entities that exist only in the organizer, never in any training text, which cleanly separates “memorized the answer” from “learned the lookup schema.”

3.3 Endpoints, statistics, and gates

The preregistration (frozen July 20, before any confirmatory run) fixes the following. H1, primary: the split arm beats its dense twin on held-out fact-use QA, split evaluated in its designed operating mode with the organizer attached, dense closed-book; the confirmation contrast at 1B uses a per-item paired bootstrap clustered by question template, with a margin of $\max(2\hat{\sigma}_{\text{pool}}, 1.0)$ points and sign consistency across seed pairs, where $\hat{\sigma}_{\text{pool}}$ is estimated from the six 160M sweep pairs. H2, guardrail: split recall with the organizer must be within two points of dense closed-book recall. H3, mediation: split store-off recall below five percent and split fact bits in weights below ten percent of the dense twin’s, so any H1 delta co-occurs with facts measurably living outside the weights. H4, secondary: fewer tokens to fixed fact-use accuracy milestones. Synthetic mathematics and deduction remain in every corpus as emergence-watch secondaries with a pre-specified reporting trigger. A defensible null is an explicit deliverable: a tight confidence interval around zero with all guardrails intact.

Three gates protected compute before the battery. Gate C required the lookup mechanics to work (they do, section 5.1). Gate B required the dose to bind on the dense arm (it does, section 5.2). Gate A required the knowledge-free tasks to be learnable at pilot budget; it failed three times at about eight GPU-hours per round, a finding in its own right (section 5.3), and triggered the endpoint re-anchoring above before any confirmatory GPU-day was spent. We consider this sequence a feature of the methodology worth emphasizing: every design risk was retired or converted into a documented decision at pilot cost, and the confirmatory battery launched with no unvalidated assumptions.

4 Experimental setup

Pilot runs train the 160M class for 0.8B tokens on a single NVIDIA L40S each (about 130k tokens per second, roughly 1.7 hours per run). The battery, running at the time of writing, comprises a dose-response sweep (three loads, two arms, two seeds, 3.2B tokens per run) and a 1B paired confirmation (two seed pairs, 10B tokens per run, dependency-chained across scheduler wall-time limits), preceded by two short 1B calibration runs whose selection rule is preregistered. All corpora were rebuilt on the frozen recipe after the gate phase. The software stack is 124 unit tests plus

Measurement	Round 1		Round 2		Round 3	
	dense	split	dense	split	dense	split
Recall, store on (%)	–	99.8	–	99.8	–	99.4
Recall, store off (%)	0.8 ^a	0.0	0.7 ^a	0.0	0.7 ^a	0.0
Fact bits in weights	40.3k	0.0	25.3k	0.0	21.3k	0.0
Fact-use QA (%)	32.9	70.6	27.1	56.9	31.2	58.1
Fresh-entity QA (%)	0.2	100.0	0.4	100.0	0.4	99.6
Malformed lookups	–	0	–	0	–	0

Table 1: Split arm versus dense twin, 200k entities, 0.8B-token pilots. Rounds differ in reasoning share and difficulty (robustness check). ^aclosed-book. Fresh-entity questions concern entities absent from all training text; 500 items per round. Fact bits estimated from attribute-level recall against pool entropies, following the accounting of [8].

an end-to-end pipeline test; corpus builds are byte-identical across serial and parallel execution; training is checkpoint-exact across requeues; every run records its configuration, git state, and full per-item evaluation outputs. Total cloud spend for everything reported here: zero dollars, on a shared university cluster.

5 Pilot results

5.1 The mechanism works, three times over

Table 1 reports the split arm and its dense twin at the 200k-entity load for each of the three pilot rounds (which differ in reasoning mixture share and difficulty band, giving an incidental robustness check). The pattern is stable everywhere it should be and null everywhere it should be.

Three results deserve emphasis. First, the separation is total: measured fact storage in the split arm’s weights is zero in every round, its store-off recall is exactly zero, and during training its cross-entropy on masked fact values stays near the uniform ceiling (about 9.5 nats against a ceiling of 10.8) while its overall loss falls normally to about 2.4. The fast and slow stores are not a metaphor in this system; they are a measured property of the trained model. Second, lookup is a learned, generalizing skill: on entities the model has never read about, it composes the correct canonical query at or near 100 percent, round after round, with zero malformed calls in roughly ten thousand attempts. Third, the behavioral payoff at the system level is large and stable: on held-out fact-use QA the split arm scores between 27 and 38 points above its dense twin, roughly doubling it, at identical size, content, and order. This is precisely the operating mode the architecture is designed for, and the margin persists across three independent trainings.

5.2 The dose binds

At fixed pilot budget the dense arm’s closed-book recall falls monotonically as the fact load rises: 1.1 percent at 50k entities, 0.8 at 200k, 0.3 at 800k. More facts strain the dense arm even at the pilot scale, through the exposure-interference mechanism the design anticipated, and the full-budget sweep quadruples exposures to size the effect properly. The 1B calibration stage selects the dose that binds at scale by a preregistered rule before the confirmation pair runs.

Round (remediation)	iGSM dense	iGSM split	Deduction dense	Deduction split
1 (share 12%, op 2–8)	2.9	4.3	44.2	47.3
2 (share 20%, op 2–6)	4.5	3.6	45.1	45.9
3 (floored: op 1–4, depth 1–2)	3.2	5.6	53.7	53.1
Chance	≈ 4.3		50.0	

Table 2: Gate A across three pilot rounds (held-out accuracy, percent, 1,500 items per cell). Knowledge-free symbolic tasks stay at chance under in-mixture exposure at 0.8B tokens and 160M parameters. Both tasks remain in the battery corpora as emergence-watch secondaries at the 4x full budget.

5.3 A clean negative: in-mixture symbolic reasoning does not ignite at pilot scale

Gate A asked whether knowledge-free mathematics and deduction, present as 12 to 20 percent of the mixture, become executable at pilot budget. Three rounds answered no (Table 2): accuracy stayed at the modular arithmetic chance floor and the binary chance line respectively, through a doubling of the reasoning share, a narrowed difficulty band, and finally a floored curriculum in which the easiest problems require a single modular operation. We read this as evidence that the arithmetic tables themselves require drilling exposure well beyond a single pass at this scale, a result consistent with the dedicated-corpus training regimes of the synthetic reasoning literature and useful to anyone designing small-model data mixtures. The gate system converted what could have been a wasted 550 GPU-hour battery into three eight-hour findings and a documented, preregistered endpoint decision.

5.4 Soundness summary

We designed for auditability and believe the pilots demonstrate it. Every contrast is a paired design isolating one causal bit. Every synthetic answer is verified by an oracle that shares no code path with generation. Held-out sets are disjoint by construction, not by filtering. The store is exact-match, so retrieval quality cannot masquerade as capacity. The mediation chain is measured at both ends: bits in weights on one side, behavioral deltas on the other. The decision rules, margins, variance estimators, exclusion policy, and endpoint hierarchy were frozen and committed before any confirmatory run started, and the one endpoint change this project has made was forced by pilot evidence, documented, and completed before the battery launched. Negative and null outcomes have preregistered, publishable forms.

6 Discussion

The pilot evidence establishes the premise of the memory-split hypothesis at small scale: it is possible, with zero training-FLOP overhead and a purely data-level intervention, to train a model whose factual knowledge lives entirely in an external, auditable, instantly editable store, while the model itself learns the general skill of consulting that store and using what it returns. The behavioral dividend on reasoning-over-facts is already visible and large in pilots. What remains, and what the running battery will decide under frozen rules, is the quantitative question at scale: how the dividend grows with fact load and whether it survives at a billion parameters with tight intervals.

For the broader research program this experiment carries a second payoff independent of the hypothesis test. The organizer, its training recipe, its interception decoder, and the accounting instru-

ments are exactly the fast-memory substrate that a fast and slow architecture needs in production: facts updatable without touching weights, deletion by row removal, provenance by construction. The science and the platform compound.

Limitations. Pilot results are at 160M parameters and 0.8B tokens; the confirmatory numbers are pending and this paper will be updated with them. The dose is synthetic; natural-corpus external validity is a designed follow-up, not a claim we make here. The fact-use endpoint compares the split system (model plus store) against a closed-book dense model, which is the deployment-relevant contrast and the preregistered one, but is a system-level rather than weights-only comparison; the knowledge-free tasks that would isolate weights-only capacity did not clear their learnability gate at pilot budget and are tracked as secondaries at full budget. Incidental facts in the natural bed are unmasked in both arms, which biases against the split arm and makes the observed margins conservative. Seed replication at 1B is two pairs, with variance estimated from six 160M pairs by a preregistered pooling rule.

7 Conclusion

A hypothesis that already steers production small-model design has been running ahead of its evidence. We built the experiment that measures it: matched twins, a controlled fact dose, an exact and auditable external memory, oracle-scored endpoints, and preregistered inference. The mechanism is proven and replicated; the behavioral separation in the architecture’s operating mode is large and stable; the negative finding en route is itself informative; and the confirmatory battery is running on frozen rules with results due within days. We believe this paradigm, validated at pilot scale for the cost of two engineer-days and zero cloud dollars, is exactly the kind of instrument that merits scaling, and we intend the next revision of this paper to carry the billion-parameter answer.

References

- [1] M. Abdin et al. Phi-3 technical report: A highly capable language model locally on your phone. arXiv:2404.14219, 2024.
- [2] L. Zhao et al. Pre-training large memory language models with internal and external knowledge. arXiv:2505.15962, 2025. ICLR 2026.
- [3] To memorize or to retrieve: Scaling laws for RAG-considerate pretraining. arXiv:2604.00715, 2026.
- [4] L. Charpentier et al. More room for language: Investigating the effect of retrieval on language models. arXiv:2404.10939, 2024.
- [5] S. Borgeaud et al. Improving language models by retrieving from trillions of tokens. arXiv:2112.04426, ICML 2022.
- [6] Great memory, shallow reasoning: Limits of kNN-LMs. arXiv:2408.11815, NAACL 2025.
- [7] V.-P. Berges, B. Oğuz, et al. Memory layers at scale. arXiv:2412.09764, 2024.
- [8] Z. Allen-Zhu and Y. Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. arXiv:2404.05405, ICLR 2025.

- [9] Z. Allen-Zhu and Y. Li (and coauthors). Physics of language models: Part 2.1, grade-school math and the hidden reasoning process. arXiv:2407.20311, ICLR 2025.
- [10] J. L. McClelland, B. L. McNaughton, and R. C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3):419–457, 1995.
- [11] D. Kumaran, D. Hassabis, and J. L. McClelland. What learning systems do intelligent agents need? *Trends in Cognitive Sciences*, 20(7):512–534, 2016.